

PAPER_900: E(t) Vs K-Essence Scherrer Model Contrast

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sion: 209 **Source:** Sessions 204-208 standalone module integration **Calculator:**
EtVsKEssenceScherrerModelContrastCalc (CP4 #484) **CVW:** v2.0.0 compliant

Abstract

Comparison of UQFF E(t) with k-essence Scherrer model dark energy. k-Essence uses non-canonical kinetic Lagrangian $F(X) = -A + BX^n$ with $X = (1/2)(\partial\phi)^2$. Derives $\rho = 2XF_X - F$, $p = F$, $w = F/(2XF_X - F)$, and sound speed $c_s^2 = F_X/(F_X + 2XF_{XX})$. 10-row contrast table covering field content, Lagrangian, EOS, sound speed, free parameters, lab testability, vacuum, GW impact, fine-tuning, and origin.

1. Core Equations

$$F(X) = -A + BX^n$$
$$w = F/(2XF_X - F)$$
$$c_s^2 = F_X/(F_X + 2XF_{XX})$$

2. Parameters

Parameter	Default	Description
A_kess	1e-47 J/m ³	k-Essence constant A
B_kess	1e-47 J/m ³	k-Essence constant B
n_kess	1.0	k-Essence power index
X_kinetic	1e-50	Kinetic term $X = (1/2)(d_\phi)^2$

3. UQFF Integration

This calculator integrates et_phonon_resonance.py from Session 208 into the CP4 pipeline as class #484.

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